Assignment 2: Instructions and Notebook Submission

Assessment 2

Deep Learning Project - Sequence Modelling



Course Code: COSC2792/COSC2779

Weighting: This assessment is worth 30% of your final mark for this Course

Length: No explicit page limit.

Task type: Individual

Due date and time: Check out the Syllabus page

Feedback: Feedback will be provided via Canvas

Click or tap on the relevant sections to learn more

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CLO1: Discuss and critically analyse a variety of neural network architectures; Evaluate and compare approaches and algorithms based on the nature of the problem/task being addressed.
- CLO2: Synthesise suitable solutions to address particular machine learning problems based on analysis of the problem and characteristics of the data involved.
- CLO3: Communicate effectively with a variety of audiences through a range of modes and media, in particular to: interpret abstract theoretical propositions, choose methodologies, justify conclusions and defend professional decisions to both IT and non-IT personnel via technical reports of professional standard and technical presentations.
- CLO4: Develop skills for further self-directed learning in the general context of neural networks and machine learning; Research, Discuss, and use new and novel algorithms for solving problems; Adapt experience and knowledge to and from other computer science contexts such as artificial intelligence, machine learning, and software design.

▼ 2. Overview of Assessment

In this assignment, you will explore a real dataset to practice the typical deep learning process. The assignment is designed to help you become more confident in applying deep learning approaches. In this assignment, you will:

- Develop a deep learning system to solve a real-world problem.
- Analyse the output of the algorithm(s).
- Research how to extend the DL techniques that are taught in class.
- Provide an ultimate judgement of the final trained model that you would use in a real-world setting.

To complete this assignment, you will require skills and knowledge from lecture and lab material for Weeks 1 to 11 (inclusive). You may find that you will be unable to complete some of the activities until you have completed the relevant lab work. However, you will be able to commence work on some sections. Thus, do the work you can initially, and continue to build in new features as you learn the relevant skills. *A deep learning model cannot be developed within a day or two. Therefore, start early.*

3. Assessment Task

3.1 The situation

You and a friend founded Litmus Pty Ltd last year to help school children understand how memes persuade them. A meme lands in under a second, and the part that does the persuading is usually the part nobody reads out loud.

In a lesson, students pull up memes from their own feeds, and you work through what each one is doing to them. Whether the meme is true is not the concern here. The question here is what technique it uses. Attaching a slur to a person. Manufacturing fear. Flattening a complicated issue into a false choice. You work to a published taxonomy of 22 persuasion techniques, and you apply it to every side of politics without exception.

The lessons work while you are in the room, because you are the one reading each meme and naming what it does. You have just signed a contract that ends that arrangement. The state education department wants the program in every school in the state, most of them taught remotely, starting next term, which is three weeks away. In those classrooms, a teacher runs the lesson using recorded material from your company, and a software needs to do the reading you used to do: a student uploads a meme, and the tool names the techniques (and where).

You have no data of your own and no time to label any. What you have is the SemEval-2021 Task 6 corpus (Dimitrov et al., 2021; cited in §3.6), a few hundred memes annotated against the same taxonomy you teach, covering 20 techniques. It is free and public.

Read the paper before you use it, where the memes were collected, when, from which groups, in whose politics and language, by how many annotators, and how those annotators settled their disagreements.

One constraint is fixed. Students upload memes from their own feeds and most of them are minors, so nothing a student enters may leave the device. No hosted model APIs, no third-party inference services.

3.2 Part A — The detector

The tool already includes a text extractor (provided; you do not build it) that reads the words off an uploaded meme and hands them to you. Your job is the classifier.

Given one meme (its extracted text) predict which of the 20 techniques it uses. A meme may use several. It may use none of the ones you predict. One meme in, a set of labels out.

The published 22 persuasion techniques are: **LABELS.txt**

You do not need to crawl social media, extract text from images, identify who appears in a meme, or judge whether anything in it is true.

Constraints

- Nothing a student enters may leave the device. You may download pretrained weights and run them locally. In deployment, you may not send a meme or its text to a hosted model API or any other third-party service.
- The tool runs on the machine the department supplies to each school, which is no better than a single NVIDIA Tesla T4 GPU with 16 GB of VRAM, and it must return an answer while a class is waiting. Keep it small enough that this is plausible, and argue that it is.
- You may use pretrained text encoders and fine-tune them. You may not use any model or checkpoint that was trained on persuasion-, propaganda- or rhetoric-labelled data.
- You may bring in additional public data to improve performance, provided you declare it explicitly in the notebook, justify it, and confirm it does not contain the labels for your evaluation set.
- Neural-network techniques only. No SVM, random forests, or other classical ML. Hand-written keyword lists or regular expressions may be used for analysis, but not as part of the final decision.

Everything else is for you to decide: how you use the meme, and how you satisfy yourself that it will hold up on whatever a Year 10 student uploads.

3.3 Part B — Showing the student where

A label on its own teaches very little. A student told that their meme uses name-calling has learned a piece of vocabulary. A student shown the three words that do the name-calling has learned to see it, which is what the lesson is for.

Extend your system to identify the span or spans of text covered by each technique. This is a multi-label sequence tagging task, an advanced form of named entity recognition, in which the same words may be covered by more than one technique at once. You may build a single model that does this and Part A together, or extend the Part A model. Either is acceptable, and your notebook should justify the choice.

3.4 Part C — The product evaluation

Before the tool goes into classrooms, the department's curriculum committee has put four questions to you. Answer each with evidence from your own system: every claim you make about your model's behaviour needs an experiment you ran and a figure or table in the notebook.

C1. Does it treat all sides the same?

The committee wants to know whether the tool is fair enough, and legible enough, to put in front of a room of teenagers who will test it on the most partisan thing in their feed.

Design and run an investigation into the biases of your system and report what you find. What to investigate is for you to determine: from the taxonomy, from where this dataset came from and how it was built, and from your own results. A bias you looked for and could not detect is worth reporting, and is marked on the quality of the test rather than on the outcome.

C2. Defend the system you built

Someone on the committee has read that general-purpose AI does this with no training at all, and wants to know why the department is paying for something you built.

Compare your system against a zero-shot general-purpose model on a representative subset of the test set, using the same measure. Making that comparison fair is most of the work: how you prompt it, how you map what it returns onto the 20 techniques, what you do with items it declines or answers in the wrong shape, and whether these public memes were plausibly in its training data.

How to run it:

- Submit the model's predictions embedded in your notebook; no need to test the full data set, a small representative sample is adequate.
- You may do the LLM calling outside the notebook;
- Record the model name and version, the date, the prompts in full, and settings such as temperature;
- Run everything downstream of those predictions, including scoring, variability and figures, in the notebook.

C3. Does the label head agree with the span head?

When the model says Name Calling and the tagger underlines nothing, the teacher has a label with nothing to point at. Measure how often your two heads disagree, per technique and overall, and say what the tool should put on screen when they do.

C4. Will it work here?

The corpus was not collected here, not recently, and not from teenagers. Your classrooms are Australian, the year is 2026, and the memes come out of a Year 10 student's own feed that week.

Identify what goes wrong in that gap and produce what evidence you can for it. There is no labelled Australian dataset waiting for you, so designing a way to get evidence at all is part of what is assessed. The gap is not only a modelling problem: consider what it means for the teacher running the lesson, and for a student whose own family's politics appear on screen with a label attached.

Constraints

- A clear, concise conclusion must be stated for each of C1 to C4.
- Every conclusion must rest on evidence from your own work: an experiment you ran, and a figure or table embedded in the notebook.
- State the limits of what your evidence can establish. A narrow conclusion that holds scores better than a broad one that reaches past its evidence.
- Report honestly where the result is unflattering. A bias you looked for and could not detect, or a baseline that outperforms your system, is a finding.
- Consider the problem broadly, including its technical, pedagogical and accountability dimensions.

3.5 Assessment Guidelines

What your notebook must document (Markdown)

You are building a deep-learning system, and every element of it must be justified using data analysis, performance analysis, and/or relevant literature (not merely asserted). Document this clearly in Markdown, at the point in the notebook where each decision is made:

- Methodology: For each element of your system, explain why it is included and the evidence supporting it.
- Evaluation framework: Explain how you selected your performance measure(s) and how you determined your data splits. Your reported performance should reflect what you can actually expect at deployment.
- Results & ultimate judgement: Analyse your model's behaviour with appropriate techniques, assess how well it performs the task in the real world, and discuss its limitations honestly.

Technical requirements

- You must use **PyTorch**, and the notebook must run on the **Colab or AWS platform** introduced in class. A notebook that does not run or is not using PyTorch scores zero.
- Your submission must execute cleanly via Restart & Run All, and must set a fixed random seed so your results are reproducible.
- All decision-supporting figures must be embedded in the notebook: learning curves, distributions, confusion matrices, and anything else you rely on to make or defend a choice. Do not depend on TensorBoard or Weights & Biases; the marker may not be able to open them.
- Neural-network techniques only. No SVM, random forests, or other classical ML.
- A clearly identifiable, self-built segment. You may use pretrained networks, but there must be network segment(s) that you designed and trained, and your code must show how that segment was developed and tuned.

3.6 Dataset

The dataset is available on Github (see below). The subset you need for the assignment is: **data_2026_A2.zip**

Dimitrov, D., Ali, B.B., Shaar, S., Alam, F., Silvestri, F., Firooz, H., Nakov, P. and Da San Martino, G., 2021, August. SemEval-2021 task 6: Detection of persuasion techniques in texts and images. In Proceedings of the 15th international workshop on semantic evaluation (SemEval-2021) (pp. 70-98).

Paper: **https://arxiv.org/pdf/2105.09284.pdf** Corpus: **https://github.com/di-dimitrov/SEMEVAL-2021-task6-corpus**

Licence agreement. The dataset can only be used for the purpose of this assignment. Sharing or distributing this data or using this data for any other commercial or non-commercial purposes is prohibited.

4. Timeline

To help you prepare for the assessment, please consider the following activities:

Timeline	Activities
Week 08	• Assignment specs will be released. • You can read the specs and start data analysis.
Week 09	• Assigment introduced in class. • Explore data and read literature that attempts to solve problems similar to the one in Assignment 2. • Establish performance measures for both the labels and the spans, and explore trivial benchmarks. • Work on the overall experimental framework design. • Start coding, data pipeline, and baseline;
Week 10	• Improve the model by reading the outputs from the model. • Run the zero-shot comparison for C2 early. It depends on an external service and everything downstream of it is blocked until you have the predictions. • Finalise both models
Week 11	• Final model, analysis, and composing the final submission. • C1, C3 and C4, then the recommendation last, once you know what the evidence says.

Timeline	Activities
	• Check your solution against the methodology in Week 6 and adjust accordingly.

5. Assessment submission and feedback

Submit your work as a Jupyter notebook via the upload link on this Canvas page. The notebook must:

- contain Part A, Part B and Part C;
- have all outputs (figures, tables, printed results) intact and visible; and
- run top-to-bottom after a kernel reset on the AWS platform without errors, if a marker chooses to re-run it.

You must also submit a PDF export of the same notebook via **Assignment 2: PDF Notebook upload** [Turnitin], which is used for similarity checking. The PDF must exactly match the submitted notebook. Any discrepancy between the two will result in a mark of zero for the assignment.

Assignments received late and without prior extension approval or special consideration will be penalised by a deduction of 10% of the total score possible per calendar day late for that assessment.

Feedback will be provided via Canvas.

6. Assessment Criteria

Your submission is marked against the three components below. The Criteria column lists the elements each component is judged on; the full grade-level descriptors (HD+, HD, DI, CR, PA, PA−, NN) for every component are given in the accompanying marking rubric at the bottom of this section.

This assessment is pitched at a high Bloom's level: **rather than following prescribed steps, you are expected to make your own reasoned judgments about how to solve this problem**. For this reason, each criterion below is stated at a high level. The detailed sub-elements the markers look for sit beneath each one, but they are deliberately not revealed at this stage. Doing so would give away the very design decisions you are being assessed on identifying. They will be shared with you alongside your feedback after marking.

Component	Criteria	Weighting
Approach (Parts A and B)	1) Data exploration leading to a well-informed approach. 2) Identifying an adequate evaluation framework tailored to the problem. 3) Well-justified network architecture and objective. 4) Hyper-parameter selection strategy. 5) Approach satisfies all the stated requirements and restrictions.	60%
Ultimate Judgment, Performance & Analysis	1) Analysis of the model and its outputs using suitable methods. 2) A clear ultimate judgment is made. 3) The rationale behind the ultimate model is clear and considers all relevant aspects. 4) Limitations of the model are identified. 5) Realistic assessment of the model's chance of success at deployment.	20%

Component	Criteria	Weighting
Part C — Product Evaluation Marked per item, C1 to C4, 1.5 marks each	1) An experiment was designed and run for each question, not asserted. 2) A supporting figure or table for each is embedded in the notebook. 3) The conclusion drawn is what the evidence supports, and its limits are stated. 4) Reporting is honest where the result is unflattering. 5) Appropriate breadth: technical, pedagogical and accountability dimensions.	20%

Detailed Rubric: [rubric-assignment2_2026.xlsx](#)

▼ 7. Academic Integrity

[Study Support Hub](#): Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

[Academic Integrity/plagiarism](#): You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

Special consideration: If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

For more information, please consult the course guide or contact your course coordinator.